Definite Integration - 1

→ Definite integration is integration with limits.

$$\int_a^b f(x)\,dx = [F(x)]_a^b = F(b) - F(a)$$

Ques $\int_3^8 \frac{\sin\sqrt{x+1}}{\sqrt{x+1}}\,dx$

$x=8 \Rightarrow t=3$, $x=3 \Rightarrow t=2$

Sol^n Let $\sqrt{x+1} = t \Rightarrow \frac{1}{2\sqrt{x+1}}\,dx = dt$

$I = \int_2^3 2\sin t\,dt \Rightarrow 2[-\cos t]_2^3$

$I = 2(\cos 2 - \cos 3)$ **Ans**

Ques $\alpha = \int_0^1 e^{9x+3\tan^{-1}x}\left(\frac{12+9x^2}{1+x^2}\right)dx$

Sol^n Let, $9x + 3\tan^{-1}x = t$ → $x=1,\ t = 9+\frac{3\pi}{4}$; $x=0,\ t=0$

$\Rightarrow \frac{dt}{dx} = 9 + 3\left(\frac{1}{1+x^2}\right) \Rightarrow \left(\frac{12+9x^2}{1+x^2}\right)dx$

$\alpha = \int_0^{9+\frac{3\pi}{4}} e^t\,dt \Rightarrow e^{(9+\frac{3\pi}{4})} - 1$ **Ans**

Ques If, $f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^2 x - 3\tan^4 x$ then find $\int_0^{\pi/4} f(x)\,dx \,/\, \int_0^{\pi/4} x f(x)$

Sol^n $f(x) = 7\tan^6 x(\tan^2 x+1) - 3\tan^2 x(\tan^2 x+1)$

$f(x) = (7\tan^6 x - 3\tan^2 x)(\sec^2 x)$

Let, $\tan x = t \Rightarrow \sec^2 x\,dx = dt$

$I_1 = \int (7t^6 - 3t^2)\,dt \Rightarrow t^7 - t^3 + C$

$I_1 = (\tan x)^7 - (\tan x)^3 + C$

$I_2 = \int x\,(7\tan^6 x - 3\tan^2 x)\,dx$ (I) (II)

$I_2 = x(\tan^7 x - \tan^3 x) - \int 1\cdot(\tan^7 x - \tan^3 x)\,dx$

$I' = \int \tan^3 x\,(\tan^4 x - 1)\cdot dx$

$I' = \int \tan^3 x\,(\tan^2 x - 1)(\tan^2 x + 1)\,dx$

$I' = \int \tan^3 x\,(\tan^2 x - 1)\sec^2 x\,dx$

[Let, $\tan x = z \Rightarrow \sec^2 x\,dx = dz$]

$I' = \int z^3(z^2-1)\,dz$

$I' = \int (z^5 - z^3)\,dz \Rightarrow \frac{z^6}{6} - \frac{z^4}{4}$

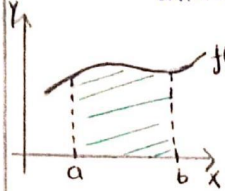
$I_2 = x(\tan^7 x - \tan^3 x) - \left[\frac{\tan^6 x}{6} - \frac{\tan^4 x}{4}\right]$

$[I_1]_0^{\pi/4} = 0$; $[I_2]_0^{\pi/2} = 1/12$

$\Rightarrow \int_0^{\pi/4} f(x)\,dx \,/\, \int_0^{\pi/4} x f(x)\,dx = 0$ **Ans**

Graphical Understanding :

$\int_a^b f(x)\,dx$ gives the area enclosed by $f(x)$ with x-axis from $x=a$ to $x=b$.



* 'Area' is always +ve

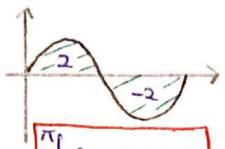
* 'Value of definite integration' is

+ve ⇒ above x-axis / right to y-axis

-ve ⇒ below x-axis / Left to y-axis

Shortcut ⇒

1) $y = \sin x$



$\int_0^{\pi} \sin x\,dx = 2$

$\int_0^{2\pi} \sin x\,dx = 0$

$\int_{\pi}^{2\pi} \sin x\,dx = -2$

2) $y = \ln x$



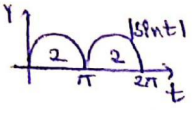
$\int_0^1 \ln x\,dx = -1$

But, area enclosed by $\ln x$ with x-axis from $x=0$ to $x=1$ is (1 units).

Ques The value of integral $\int_0^{\pi} |\sin 2x|\,dx$ is ____

Sol^n Let, $2x = t$, $2\,dx = dt$; $x=0,\ t=0$; $x=\pi,\ t=2\pi$

$I = \frac{1}{2}\int_0^{2\pi} |\sin t|\,dt$



$I = \frac{4}{2} = 2$ **Ans**

Properties of Definite Integration :

* **Property-01 ⇒** $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$

Integrating variable can be changed.

* **Property-02 ⇒** $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

Minus is introduced when limits are reversed.

* **Property-03 ⇒** $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

Use this property when integral is broken at points of discontinuity or at the points where definition of $f(x)$ changes.

Eg: $\int_0^3 [x]\,dx$

Algebraic Method ⇒

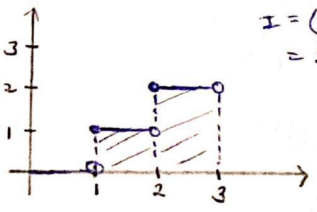
Break the function at the points where its value changes, and also substitute values of $[x]$ according to limits of integration.

$I = \int_0^1 0\cdot dx + \int_1^2 1\cdot dx + \int_2^3 2\cdot dx$

$I = 0 + (x)_1^2 + 2(x)_2^3 \Rightarrow 1 + 2$

$I = 3$ **Ans**

Graphical Method ⇒



$I = (1\times1) + (2\times1)$

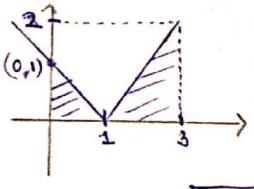
$= 1+2 \Rightarrow 3$ **Ans**

Eg: $\int_0^3 |x-1|\,dx$

When, $x=0$, $y = |x-1| = 1$

$x=3$, $y = |3-1| = 2$

Sol^n



$I = \left(\frac{1}{2}\times1\times1\right) + \left(\frac{1}{2}\times2\times2\right)$

$\Rightarrow I = \frac{1}{2} + 2 \Rightarrow \frac{5}{2}$ **Ans**

Ques Find value of $\int_{-1}^{1} x^3\cdot e^{[x^3]}\,dx$

Sol^n

$I = \int_{-1}^{0} x^3\cdot e^{[x^3]}\,dx + \int_0^1 x^3\cdot e^{[x^3]}\,dx$

$-1 < x < 0$, $-1 < x^3 < 0$ ⇒ $[x^3] = -1$

$0 < x < 1$, $0 < x^3 < 1$ ⇒ $[x^3] = 0$

$I = \int_0^1 x^3\cdot e^0\,dx + \int_{-1}^0 x^3\cdot e^{-1}\,dx$

$I = \frac{x^4}{4}\Big|_0^1 + \frac{1}{e}\times\frac{x^4}{4}\Big|_{-1}^0$

$I = \frac{1}{4} + \left(-\frac{1}{4e}\right) \Rightarrow I = \frac{e-1}{4e}$ **Ans**

* **Property-04 ⇒**

$$\int_{-a}^{a} f(x)\,dx = \int_0^a \left(f(x)+f(-x)\right)dx = \begin{cases} 0 ; & \text{if } f(x) = \text{odd} \\ 2\int_0^a f(x)\,dx ; & \text{if } f(x) = \text{even} \end{cases}$$

If $f(-x) = -f(x) \Rightarrow f(x)$ is odd

$f(-x) = f(x) \Rightarrow f(x)$ is even

Use this property only when limits are conjugate of each other i.e. a & $-a$.

Ques $\int_{-1}^{1} x^3 \tan(x^2)\,dx$

Sol^n $f(-x) = (-x)^3 \tan[(-x)^2]$

$= -x^3 \tan(x^2) \Rightarrow$ Odd function

So, $\boxed{\int_{-1}^{1} x^3 \tan(x^2)\cdot dx = 0}$ Ans

Ques $\int_{-\pi/2}^{\pi/2} \sec x \cdot \ln\left(\frac{1-x}{1+x}\right)\cdot dx$

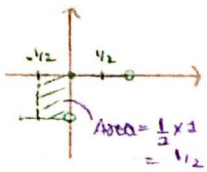
Sol^n $f(-x) = \sec(-x)\cdot \ln\left(\frac{1+x}{1-x}\right)$

$= -\sec x \cdot \ln\left(\frac{1-x}{1+x}\right) = -f(x) \Rightarrow$ odd function

So, $\boxed{\int_{-\pi/2}^{\pi/2} f(x)\cdot dx = 0}$ Ans

Ques $\int_{-1/2}^{1/2} \left([x] + \ln\left(\frac{1+x}{1-x}\right)\right) dx$

Sol^n $\int_{-1/2}^{1/2} [x]\cdot dx + \int_{-1/2}^{1/2} \ln\left(\frac{1+x}{1-x}\right)\cdot dx$ (0 (odd func^n))



$\boxed{\int_{-1/2}^{1/2} [x]\cdot dx = -\frac{1}{2}}$ Ans

Ques $\int_{-\pi/2}^{\pi/2} \left(x^2 + \ln\left(\frac{\pi - x}{\pi + x}\right)\right)\cos x \cdot dx$

Sol^n $\int_{-\pi/2}^{\pi/2} x^2\cdot\cos x\,dx + \int_{-\pi/2}^{\pi/2} \cos x \cdot \ln\left(\frac{\pi-x}{\pi+x}\right)\cdot dx$ (0 (odd func^n))

(even func^n)

D	I
$\oplus$ x^2	$\cos x$
$\ominus$ $2x$	$\sin x$
$\oplus$ 2	$-\cos x$
$\ominus$ 0	$-\sin x$

$\Rightarrow 2\int_0^{\pi/2} x^2\cdot\cos x\cdot dx$

$\Rightarrow 2\int_0^{\pi/2} x^2\cdot\cos x\cdot dx = 2\left[x^2\sin x + 2x\cos x - 2\sin x\right]_0^{\pi/2}$

$= 2\left[\frac{\pi^2}{4} - 2\right] \Rightarrow \boxed{\frac{\pi^2}{2} - 4}$ Ans

Imp

* Property - 05 :-

$$\int_a^b f(x)\cdot dx = \int_a^b f(a+b-x)\cdot dx$$

Shortcut $\Rightarrow$

Let, $I_1 = \int_a^b f(x)\cdot dx$ and $\int_a^b f(a+b-x)\cdot dx$.

If, on adding both I_1 & I_2, we get $I_1 + I_2 = \int_a^b 1\cdot dx$ then we can directly write

$$I_1 = \int_a^b f(x)\cdot dx = \frac{b-a}{2}$$

Ques $\int_{50}^{100} \frac{\ln x}{\ln x + \ln(150-x)}\cdot dx$

Sol^n $f(x) = \frac{\ln x}{\ln x + \ln(150-x)}$

$f(a+b-x) = \frac{\ln(150-x)}{\ln(150-x) + \ln x}$

$f(x) + f(a+b-x) = \frac{\ln x + \ln(150-x)}{\ln x + \ln(150-x)} = 1$

So, $\int_{50}^{100} f(x)\cdot dx = \frac{100-50}{2} = \boxed{25}$ Ans

Ques $\int_2^3 \frac{\sqrt{5-x}}{\sqrt{x} + \sqrt{5-x}}\,dx$

Sol^n $f(a+b-x) = \frac{\sqrt{x}}{\sqrt{5-x} + \sqrt{x}}$

$f(x) + f(a+b-x) = \frac{\sqrt{5-x} + \sqrt{x}}{\sqrt{5-x} + \sqrt{x}} \Rightarrow 1$

So, $\int_2^3 f(x)\cdot dx = \frac{3-1}{2} = \boxed{\frac{1}{2}}$ Ans

Shortcut $\Rightarrow$

In definite integration we can directly replace, terms like

$\frac{1}{1+e^x}$ / $\frac{1}{1+a^x}$ / $\frac{1}{1+a^{\sin x}}$ / $\frac{1}{1+a^{\tan x}}$

by $\left(\frac{1}{2}\right)$

Ques $\int_0^{\pi} \frac{dx}{1 + 2^{\tan x}}$

Sol^n $I = \int_0^{\pi} \frac{dx}{1 + 2^{\tan(\pi - x)}} \Rightarrow \int_0^{\pi} \frac{dx}{1 + 2^{-\tan x}}$

$I = \int_0^{\pi} \frac{2^{\tan x}}{1 + 2^{\tan x}}\cdot dx$

Now,

$I + I = \int_0^{\pi} \frac{1 + 2^{\tan x}}{1 + 2^{\tan x}}\cdot dx \Rightarrow \int_0^{\pi} dx$

$2I = \pi \Rightarrow \boxed{I = \pi/2}$ Ans

Ques $\int_{-\pi/2}^{\pi/2} \frac{1}{1 + e^{\sin x}}\cdot dx$

Sol^n $I = \int_{-\pi/2}^{\pi/2} \left(\frac{1}{2}\right)\cdot dx \Rightarrow \frac{1}{2}\left(\frac{\pi}{2} + \frac{\pi}{2}\right)$

$\Rightarrow \boxed{\pi/2}$ Ans

Ques $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1 + a^x}\cdot dx$

Sol^n $I = \frac{1}{2}\int_{-\pi}^{\pi} \cos^2 x\cdot dx \Rightarrow \frac{1}{2}\int_0^{\pi} \cos^2 x\cdot dx$

$I = \int_0^{\pi} \frac{1 + \cos 2x}{2}\cdot dx \Rightarrow \left[\frac{x}{2} + \frac{\sin 2x}{4}\right]_0^{\pi}$

$\boxed{I = \frac{\pi}{2}}$ Ans

Ques If $I = \frac{2}{\pi}\int_{-\pi/4}^{\pi/4} \frac{dx}{(1+e^{\sin x})(2-\cos 2x)}$ then find $27I^2$.

Sol^n $I = \frac{2}{\pi} \times \frac{1}{2}\int_{-\pi/4}^{\pi/4} \frac{dx}{2 - (2\cos^2 x - 1)}$

Since $f(x)$ is even now, so we can write

$I = \frac{2}{\pi}\int_0^{\pi/4} \frac{dx\cdot \sec^2 x}{(3 - 2\cos^2 x)\sec^2 x}$

$I = \frac{2}{\pi}\int_0^{\pi/4} \frac{dx\cdot \sec^2 x}{3(1 + \tan^2 x) - 2}$

Let, $\tan x = t \Rightarrow \sec^2 x\,dx = dt$

Also, $x = \pi/4 \Rightarrow \tan x = 1$

$x = 0 \Rightarrow \tan x = 0$

So, $I = \frac{2}{\pi}\int_0^1 \frac{dt}{3(1+t^2) - 2}$

$I = \frac{2}{\pi}\int_0^1 \frac{dt}{3t^2 + 1} \Rightarrow \frac{2}{3\pi}\int_0^1 \frac{dt}{t^2 + \left(\frac{1}{\sqrt{3}}\right)^2}$

$I = \frac{2}{3\pi} \times \frac{1}{\left(\frac{1}{\sqrt{3}}\right)} \times \left[\tan^{-1}\left(\frac{t}{1/\sqrt{3}}\right)\right]_0^1$

$I = \frac{2\sqrt{3}}{3\pi}\left(\pi/3\right) \Rightarrow \boxed{I = \frac{2}{3\sqrt{3}}}$

So, $\boxed{27I^2 = 27 \times \frac{4}{27} = 4}$ Ans

* Property - 06 $\Rightarrow$

$$\int_0^{2a} f(x)\cdot dx = \int_0^a f(x)\cdot dx + \int_0^a f(2a-x)\cdot dx$$

We will use this property only when lower limit is zero.

Ques $I = \int_0^{2\pi} \sin^4 x\cdot dx = K\int_0^{\pi/2} \cos^4 x\cdot dx$, find K.

Sol^n $I = \int_0^{2\pi} \sin^4 x\cdot dx \Rightarrow \int_0^{\pi}\left(\sin^4 x + \sin^4(2\pi - x)\right) dx$

$I = \int_0^{\pi}(\sin^4 x + \sin^4 x)\,dx \Rightarrow 2\int_0^{\pi} \sin^4 x$

$I = 2\int_0^{\pi/2}(\sin^4 x + \sin^4(\pi - x))\,dx \Rightarrow 2\int_0^{\pi/2}(\sin^4 x + \sin^4 x)\,dx$

$I = 4\int_0^{\pi/2} \sin^4 x\,dx \Rightarrow 4\int_0^{\pi/2} \sin^4\left(\frac{\pi}{2} + 0 - x\right) dx$

$I = 4\int_0^{\pi/2} \cos^4 x\cdot dx \Rightarrow \boxed{K = 4}$ Ans

Definite Integration - 2

Ques If $\int_0^{\pi} x\, f(\sin x)\,dx = A\int_0^{\frac{\pi}{2}} f(\sin x)\,dx$, find A.

Sol^n 'x' ko product mein se hatane ke liye, we always need to use property-5 i.e. $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

$$I = \int_0^{\pi} x\cdot f(\sin x)\,dx \quad -(1)$$

$$I = \int_0^{\pi} (\pi - x)\cdot f[\sin(\pi - x)]\,dx$$

$$I = \int_0^{\pi} \pi\cdot f(\sin x)\,dx - \int_0^{\pi} x\, f(\sin x)\,dx$$

$$\Rightarrow \quad 2I = \int_0^{\pi} \pi\, f(\sin x)\,dx \quad (\text{By eqn } (1))$$

$$2I = \pi\int_0^{\pi} \left(f(\sin x) + f(\sin(\pi - x))\right)dx$$

$$2I = \pi\int_0^{\pi} \left(f(\sin x) + f(\sin x)\right)dx$$

$$2I = 2\pi\int_0^{\pi/2} f(\sin x)\,dx \Rightarrow \boxed{A = \pi}$$ Ans

Important Result ⇒

$$\int_0^{\pi/2} \ln(\sin x)\,dx = \int_0^{\frac{\pi}{2}} \ln(\cos x)\,dx = \int_0^{\pi/2} \ln(\sin 2x)\,dx = \frac{-\pi}{2}\ln 2$$

*** Property-07 ⇒**

If $f(x)$ is a periodic function of period T then

$$\int_0^{nT} f(x)\,dx = n\int_0^{T} f(x)\,dx$$

Ques $\int_0^{1000} e^{x-[x]}\,dx$

Sol^n $\because\ x - [x] = \{x\}$

$$I = \int_0^{1000} e^{\{x\}}\,dx$$ ← Repeats after 1

$[\because \text{in } (0,1),\ \{x\} = x \text{ only}]$

$$I = 1000\int_0^{1} e^{\{x\}}\,dx \Rightarrow 1000\int_0^{1} e^{x}\,dx$$

$$\boxed{I = 1000(e-1)}$$ Ans

Ques Let $f(x)$ be a continuous periodic function such that $f(x+T) = f(x)$. If $I = \int_0^T f(x)\,dx$ then find $\int_3^{3+3T} f(2x)\,dx$ in terms of I.

Sol^n For, $f(x) \longrightarrow$ Period $= T$

$f(2x) \longrightarrow$ Period $= T/2$

So, No. of cycles $(n) = \frac{3+3T-3}{T/2} \Rightarrow \boxed{n = 6}$

Again,

$$\int_3^{3+3T} f(2x)\,dx = 6\int_0^{T/2} f(2x)\,dx \quad \left[\text{let, } 2x = t,\ dx = \frac{dt}{2}\right]$$

$$= \frac{6}{2}\int_0^{T} f(t)\,dt \Rightarrow 3\int_0^{T} f(x)\,dx \quad (\text{Property-1})$$

$$\boxed{\int_3^{3+3T} f(2x)\,dx = 3I}$$ Ans

Shortcuts for periodic Function ⇒

1) We can calculate no. of cycles by the expression

$$n = \frac{(\text{Upper limit}) - (\text{Lower limit})}{\text{Time Period}}$$

2) Following are direct application of Property-07 ⇒

(i) $\int_a^{a+nT} f(x)\,dx = n\int_0^{T} f(x)\,dx$

(ii) $\int_{mT}^{nT} f(x)\,dx = (n-m)\int_0^{T} f(x)\,dx$

(iii) $\int_{hT}^{a+hT} f(x)\,dx = \int_0^{a} f(x)\,dx$ ← If 'nT' is present in both up & low limit, then it can be deleted

(iv) $\int_{a+nT}^{b+nT} f(x)\,dx = \int_a^{b} f(x)\,dx$

Leibniz Rule :

If f is continuous then

$$\frac{d}{dx}\left(\int_{g(x)}^{h(x)} f(t)\,dt\right) = f(h(x))\cdot h'(x) - f(g(x))\cdot g'(x)$$

Ques $\lim_{x\to 0} \frac{\int_0^{x^2} (\sin\sqrt{t})\,dt}{x^3}$ $\left(\frac{0}{0}\text{ form}\right)$

Sol^n We need to apply LH rule, so

$$K = \lim_{x\to 0} \frac{\frac{d}{dx}\sin\sqrt{x^2}\cdot(2x) - \sin\sqrt{0}\cdot(0)}{3x^2}$$

$$K = \lim_{x\to 0} \frac{2\sin x}{3x} \Rightarrow \boxed{\frac{2}{3}}$$ Ans

Definite Integeration as Limit of Sum :

Shortcut ⇒

Replace $\sum \to \int$, $\frac{r}{n} \to x$ and $\frac{1}{n} \to dx$

Ques $\lim_{n\to\infty}\left(\frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \cdots + \frac{1}{5n}\right)$ is equal to...

Sol^n

$$= \lim_{n\to\infty}\left(\sum_{r=1}^{2n} \frac{n}{n^2+r^2}\right) \Rightarrow \lim_{n\to\infty}\left(\sum_{r=1}^{2n} \frac{\frac{1}{n}}{\left(1+\frac{r^2}{n^2}\right)n}\right)$$

$$= \lim_{n\to\infty} \sum_{r=1}^{2n} \frac{1}{n\left(1+\left(\frac{r}{n}\right)^2\right)} \Rightarrow \int_0^2 \frac{1}{1+x^2}\,dx$$

$$\left[\begin{array}{l} 1 \le r \le 2n \\ \frac{1}{n} \le \frac{r}{n} \le 2 \Rightarrow 0 \le x \le 2 \\ \qquad\quad (\text{Because } n\to\infty) \end{array}\right]$$

$$= \boxed{\tan^{-1}(2)}$$ Ans

Ques Let, $Y_n = \frac{1}{n}\left[(n+1)(n+2)\cdots(2n)\right]^{1/n}$ and $\lim_{n\to\infty} Y_n = L$, then find $[L]$.

Sol^n Whenever terms are in product, take log both sides.

$$Y_n = \left[\frac{(n+1)(n+2)\cdots(2n)}{n^n}\right]^{1/n} \Rightarrow \left[\left(\frac{n+1}{n}\right)\left(\frac{n+2}{n}\right)\cdots\left(\frac{2n}{n}\right)\right]^{1/n}$$

$$\ln y = \frac{1}{n}\left[\ln\left(\frac{n+1}{n}\right) + \ln\left(\frac{n+2}{n}\right) \cdots + \ln\left(\frac{n+n}{n}\right)\right]$$

$$\ln y = \frac{1}{n}\left(\sum_{r=1}^{n} \ln\left(\frac{n+r}{n}\right)\right)$$

$$\lim_{n\to\infty} \ln y = \lim_{n\to\infty} \frac{1}{n}\sum_{r=1}^{n} \ln\left(1+\frac{r}{n}\right)$$

$$\ln y = \int_0^1 \ln(1+x)\,dx \quad \left[\begin{array}{l} 1 \le r \le n \\ \frac{1}{n} \le \frac{r}{n} \le 1 \\ 0 \le x \le 1 \end{array}\right]$$

$$\ln y = \ln(1+x)\cdot x - \int \frac{x}{1+x}\,dx$$

$$= \ln(1+x)\cdot x - \int\left(\frac{x+1-1}{1+x}\right)dx$$

$$= x\cdot\ln(x+1) - \int dx + \int \frac{1}{1+x}\,dx$$

$$\ln y = \left[x\ln(x+1) - x + \ln(1+x)\right]_0^1$$

$$\ln y = \ln 2 - 1 + \ln 2 \Rightarrow 2\ln 2 - \ln e$$

$$\ln y = \ln\left(\frac{4}{e}\right) \Rightarrow \boxed{y = \frac{4}{e}}$$

i.e. $L = \frac{4}{2.71} = 1.\overline{mm} \Rightarrow \boxed{[L] = 1}$ Ans

General Inequalities in Definite Integration :

→ Some integrals can't be calculated in mathematics.

$$\int \frac{\sin x}{x}\,dx \quad \int \frac{\cos x}{x}\,dx \quad \int \sqrt{\sin x}\,dx \quad \int \sin x^2\,dx$$

$$\int \cos x^2\,dx \quad \int x\cdot\tan x\,dx \quad \int e^{-x^2}\,dx \quad \int e^{x^3}\,dx$$

$$\int \frac{x^3}{1+x^5}\,dx \quad \int (1+x^2)^{1/3}\,dx \quad \int \frac{dx}{\ln x}$$

$$\int \left(\sqrt{1+k^2\sin^2 x}\right)dx$$

→ So, instead of calculating exact values of these Integrals, we take help of some Inequalities.

(I) Rectangle Method :→

a) For a monotonic rising function ⇒

$$(b-a)f(a) < \int_a^b f(x)\,dx < (b-a)f(b)$$

(b) For a monotonic falling function ⇒

$$f(b)\cdot(b-a) < \int_a^b f(x)\,dx < (b-a)f(a)$$

(c) For non-monotonic function in (a,b) ⇒

$$m\cdot(b-a) < \int_a^b f(x)\,dx < M\cdot(b-a)$$

where,

m → Min. value of f(x) in (a,b)

M → Max. value of f(x) in (a,b)

NOTE:

$$\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx$$

Equality holds only when f(x) completely lies above x-axis.

(II) Use of Known functions :→

If $h(x) \le f(x) \le g(x)\ \forall\ x \in [a,b]$

then $\int h(x)\,dx \le \int f(x)\,dx \le \int g(x)\,dx$

Wallis Theorem :

$$\int_0^{\pi/2} \sin^n x\cdot\cos^m x\,dx = \frac{[(n-1)(n-3)\cdots 1/2][(m-1)(m-3)\cdots 1/2]}{(m+n)(m+n-2)\cdots 2\,1/2} \times K$$

where, $K = \begin{cases} \pi/2, & \text{if } m,n \text{ both even} \\ 1, & \text{Rest all case} \end{cases}$

* Keep substracting until 1 or 2 is obtained
* Applicable only when limits → $[0, \pi/2]$

Ques $\int_0^{\pi/2} \sin^6 x\cdot\cos^4 x\,dx$

Sol'n

$$I = \frac{[(6-1)(6-3)(6-5)]\times[(4-1)(4-3)]}{(10)(10-2)(10-4)(10-6)(10-8)} \times \frac{\pi}{2}$$

$$I = \frac{(5\times3\times1)\times(3\times1)}{10\times8\times6\times4\times2} \times \frac{\pi}{2}$$

$$I = \frac{3\pi}{2^9}$$ Ans

Ques $\int_0^{\pi/2} \sin^5 x\,dx$

Sol'n

$$I = \int_0^{\pi/2} \sin^5 x\cdot\cos^0 x\,dx$$

$$I = \frac{(5-1)(5-3)}{(5)(5-2)(5-4)} \times 1$$

$$I = \frac{4\times2}{5\times3\times1} \Rightarrow \frac{8}{15}$$ Ans

Ques $\int_{-\pi/2}^{\pi/2} \sin^2 x\cdot\cos^2 x(\sin x+\cos x)\,dx$

Sol'n

$$I = \int_{-\pi/2}^{\pi/2} \sin^3 x\cdot\cos^2 x\,dx + \int_{-\pi/2}^{\pi/2} \sin^2 x\cdot\cos^3 x\,dx$$

(odd → 0; even)

$$I = 2\int_0^{\pi/2} \sin^2 x\cdot\cos^3 x\,dx$$

$$I = 2\times\frac{(2-1)\times(3-1)}{(5)(5-2)(5-4)} \times 1$$

$$I = 2\times\frac{1\times2}{5\times3\times1} \times 1$$

$$I = \frac{4}{15}$$ Ans